Dynamic Frequency Pairing

Time Limit: 3.0 seconds
Memory Limit: 512 MB

You are given an array A of N integers (1-indexed). You need to process Q dynamic queries of two types on this array:

- Type 1 (`1 i x`): Update the element at index i to value x (i.e., set $A_i = x$).
- Type 2 (`2 L R P Q`): Consider the contiguous subarray A[L … R]. Select two distinct values u and v ($u \neq v$) present in A[L … R] such that:
  - $freq(u) \geq P$ and $freq(v) \geq Q$, OR
  - $freq(u) \geq Q$ and $freq(v) \geq P$.

Here, freq(x) denotes the number of occurrences of value x within the subarray A[L … R].

The score for a valid pair (u, v) is defined as:

$$Score = (u \times freq(u)) + (v \times freq(v))$$

Output the maximum possible score achievable by any valid pair (u, v) in the subarray A[L … R]. If no such pair of distinct values exists, output 0.

Input

- The first line contains two integers N and Q ($2 \leq N \leq 10^5$, $1 \leq Q \leq 10^5$) — the size of the array and the number of queries.
- The second line contains N space-separated integers $A_1, A_2, \ldots, A_N$ ($1 \leq A_i \leq 10^9$) — the initial elements of the array.
- The next Q lines each contain a query in one of two formats:
  - `1 i x` ($1 \leq i \leq N$, $1 \leq x \leq 10^9$)
  - `2 L R P Q` ($1 \leq L \leq R \leq N$, $1 \leq P, Q \leq R - L + 1$)

Output

For each query of Type 2, print a single integer on a new line representing the maximum score obtainable, or 0 if no two distinct elements satisfy the frequency constraints.

Constraints

- $2 \leq N \leq 10^5$
- $1 \leq Q \leq 10^5$
- $1 \leq A_i, x \leq 10^9$
- $1 \leq L \leq R \leq N$
- $1 \leq P, Q \leq R - L + 1$

Sample Input

```
6 4
2 2 2 5 5 3
2 1 6 2 3
1 6 5
2 1 6 3 3
2 4 6 2 2
```

Sample Output

```
16
21
0
```

Explanation of Sample

- Initial Array: A = [2, 2, 2, 5, 5, 3]
- Query 1 (`2 1 6 2 3`):
  - Subarray A[1 … 6] = [2, 2, 2, 5, 5, 3].
  - Frequencies: freq(2) = 3, freq(5) = 2, freq(3) = 1.
  - Require one frequency ≥ 3 and one frequency ≥ 2.
  - Selecting u = 2 (freq = $3 \geq 3$) and v = 5 (freq = $2 \geq 2$) satisfies the condition.
  - Score = $(2 \times 3) + (5 \times 2) = 6 + 10 = 16$.
- Query 2 (`1 6 5`):
  - Update $A_6 = 5$.
  - Array becomes A = [2, 2, 2, 5, 5, 5].
- Query 3 (`2 1 6 3 3`):
  - Subarray A[1 … 6] = [2, 2, 2, 5, 5, 5].
  - Frequencies: freq(2) = 3, freq(5) = 3.
  - Both thresholds require frequency ≥ 3.
  - Pair (2, 5) satisfies this constraint.
  - Score = $(2 \times 3) + (5 \times 3) = 6 + 15 = 21$.
- Query 4 (`2 4 6 2 2`):
  - Subarray A[4 … 6] = [5, 5, 5].
  - Frequencies: freq(5) = 3.
  - There is only 1 distinct element in this range, but the problem requires selecting two distinct values ($u \neq v$). No valid pair exists, so the output is 0.